

CIL MT 2026 — FULL-LENGTH MOCK CBT 2

Coal India Limited | Management Trainee | System Discipline (Post Code 14) | Advt. No. 03/2026

Total questions: 200 | **Total marks:** 200 | **Total time:** 3 hours (180 minutes), single sitting

Paper-I — General Knowledge/Awareness, Reasoning, Numerical Ability, General English	Q1 – Q100 (100 marks)
Paper-II — Professional Knowledge, System (CS/IT)	Q101 – Q200 (100 marks)

Instructions — follow these exactly to make the practice count

1. Set a single timer for 180 minutes and do not pause it. Both papers are attempted in one sitting, exactly as in the CBT.
 2. Each question carries 1 mark. **There is no negative marking** — leave nothing blank. Mark your best guess on anything you cannot solve.
 3. Both papers must be qualified **separately**: 40 marks for UR/EWS, 35 for OBC-NCL, 30 for SC/ST/PwD, out of 100 in each paper. Score each paper on its own before looking at the total.
 4. Suggested split: 75 minutes on Paper-I, 95 minutes on Paper-II, 10 minutes to sweep unattempted questions. Adjust after your first attempt.
 5. The answer key is at the end of this document. Do not open it until the timer stops.
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PAPER-I — GENERAL APTITUDE (Q1 - Q100)

SECTION A — GENERAL KNOWLEDGE / AWARENESS

- Q1.** The SI unit of pressure is the
(A) watt (B) pascal (C) newton (D) joule
- Q2.** Gondwana coal deposits account for approximately what share of India's total coal reserves?
(A) 30% (B) 98% (C) 50% (D) 75%
- Q3.** The Jharia coalfield, famous for its coking coal reserves, is located in
(A) Odisha (B) West Bengal (C) Chhattisgarh (D) Jharkhand
- Q4.** Which variety of coal has the highest carbon content and calorific value?
(A) Lignite (B) Bituminous (C) Anthracite (D) Peat
- Q5.** The Comptroller and Auditor General of India is appointed under which Article?
(A) Article 315 (B) Article 76 (C) Article 324 (D) Article 148
- Q6.** Which state possesses the largest coal reserves in India?
(A) Odisha (B) Chhattisgarh (C) West Bengal (D) Jharkhand
- Q7.** President's Rule in a state is imposed under which Article?
(A) Article 356 (B) Article 352 (C) Article 360 (D) Article 370
- Q8.** Deficiency of vitamin C causes
(A) scurvy (B) beriberi (C) rickets (D) night blindness
- Q9.** Coal India Limited was conferred Maharatna status in the year
(A) 2008 (B) 2014 (C) 2011 (D) 2010
- Q10.** The Talcher coalfield lies in the state of
(A) Jharkhand (B) Telangana (C) Odisha (D) Chhattisgarh
- Q11.** The 73rd Constitutional Amendment Act is associated with
(A) anti-defection (B) the right to education (C) urban local bodies (D) Panchayati Raj institutions
- Q12.** The chemical formula of baking soda is
(A) NaHCO_3 (B) CaCO_3 (C) NaCl (D) Na_2CO_3
- Q13.** Which of the following has the lowest carbon content?
(A) Anthracite (B) Bituminous (C) Lignite (D) Peat
- Q14.** Which Indian state is the leading producer of lignite?
(A) Tamil Nadu (B) Rajasthan (C) Gujarat (D) Jharkhand
- Q15.** The Battle of Plassey was fought in the year
(A) 1764 (B) 1782 (C) 1757 (D) 1857
- Q16.** The largest gland in the human body is the
(A) pancreas (B) liver (C) thyroid (D) pituitary
- Q17.** The longest river flowing entirely within Indian territory is the
(A) Ganga (B) Godavari (C) Narmada (D) Brahmaputra
- Q18.** Article 32 of the Constitution of India provides for
(A) the abolition of untouchability (B) constitutional remedies through writs (C) the right to property (D) the right to freedom of religion
- Q19.** The Jallianwala Bagh massacre took place in the year
(A) 1930 (B) 1922 (C) 1909 (D) 1919
- Q20.** The largest state of India by geographical area is
(A) Rajasthan (B) Madhya Pradesh (C) Uttar Pradesh (D) Maharashtra
- Q21.** CMPDIL, a subsidiary of Coal India Limited, is primarily engaged in
(A) coal production (B) power generation (C) exploration, planning and consultancy for the coal sector (D) coal transportation
- Q22.** Western Coalfields Limited (WCL) is headquartered at
(A) Pune (B) Bhopal (C) Nagpur (D) Raipur
- Q23.** Bharat Coking Coal Limited (BCCL) has its headquarters at
(A) Bokaro (B) Dhanbad (C) Ranchi (D) Asansol
- Q24.** Kaziranga National Park is chiefly known for the
(A) snow leopard (B) Bengal tiger only (C) Asiatic lion (D) one-horned rhinoceros
- Q25.** Who among the following is known as the "Iron Man of India"?
(A) Sardar Vallabhbhai Patel (B) Subhas Chandra Bose (C) Lala Lajpat Rai (D) Bal Gangadhar Tilak

SECTION B — REASONING

- Q26.** Fish : Gills :: Human : ?
(A) Lungs (B) Skin (C) Liver (D) Heart
- Q27.** A man walks 3 km east and then 4 km north. His distance from the starting point is
(A) 5 km (B) 1 km (C) 7 km (D) 12 km

Q28. Choose the odd one out.

(A) 200 (B) 121 (C) 144 (D) 169

Q29. Statements: All cats are animals. All animals have tails. Which conclusion necessarily follows?

(A) All animals are cats (B) Some cats have no tails (C) All cats have tails (D) No conclusion follows

Q30. Thermometer : Temperature :: Barometer : ?

(A) Wind speed (B) Pressure (C) Rainfall (D) Humidity

Q31. A person facing north turns 90° clockwise twice. He is now facing

(A) south (B) west (C) north (D) east

Q32. A is taller than B but shorter than C. Who is the tallest?

(A) B (B) C (C) A (D) Cannot be determined

Q33. Find the next term: 7, 14, 28, 56, ?

(A) 98 (B) 112 (C) 84 (D) 126

Q34. Find the next term: 3, 9, 27, 81, ?

(A) 324 (B) 162 (C) 729 (D) 243

Q35. Find the next term: 81, 64, 49, 36, ?

(A) 16 (B) 27 (C) 25 (D) 30

Q36. A is to the right of B, and C is to the right of A. Who is in the middle?

(A) B (B) C (C) A (D) Cannot be determined

Q37. A is the sister of B, and B is the son of C. How is A related to C?

(A) Daughter (B) Sister (C) Mother (D) Niece

Q38. If LIGHT is coded as MJHIU, then DARK is coded as

(A) CZQJ (B) EBSK (C) EBSL (D) EBTL

Q39. Complete the series: B, D, G, K, ?

(A) N (B) O (C) Q (D) P

Q40. In a queue, a person is 7th from the front and 9th from the back. The total number of persons in the queue is

(A) 16 (B) 14 (C) 17 (D) 15

Q41. If MOTHER is coded as NPUIFS, then FATHER is coded as

(A) GBTIFS (B) GBUIFT (C) EZSGDQ (D) GBUIFS

Q42. Complete the series: A, C, F, J, O, ?

(A) U (B) T (C) V (D) S

Q43. Choose the odd one out.

(A) Cube (B) Pentagon (C) Hexagon (D) Triangle

Q44. Complete the series: Z, X, V, T, ?

(A) S (B) P (C) R (D) Q

Q45. Find the next term: 5, 11, 23, 47, ?

(A) 95 (B) 94 (C) 96 (D) 71

Q46. Statements: All pens are books. Some books are red. Which conclusion necessarily follows?

(A) All books are pens (B) Some pens are red (C) No pen is red (D) None of these follows

Q47. 4 : 16 :: 5 : ?

(A) 25 (B) 20 (C) 30 (D) 36

Q48. If A = 1, B = 2, C = 3 and so on, the sum of the letters of the word CAB is

(A) 6 (B) 5 (C) 7 (D) 9

Q49. In a class of 30 students, a boy ranks 8th from the top. His rank from the bottom is

(A) 23rd (B) 24th (C) 21st (D) 22nd

Q50. If 1 January 2024 was a Monday, then 1 January 2023 was a

(A) Monday (B) Tuesday (C) Saturday (D) Sunday

SECTION C — NUMERICAL ABILITY

Q51. 60 is what percentage of 480?

(A) 15% (B) 10% (C) 8% (D) 12.5%

Q52. Two numbers are in the ratio 2 : 3 and their difference is 12. The larger number is

(A) 30 (B) 36 (C) 24 (D) 48

Q53. The sum of the ages of 5 family members is 100 years. Their average age is

(A) 22 years (B) 25 years (C) 20 years (D) 18 years

Q54. The percentage increase when a quantity rises from 40 to 50 is

(A) 10% (B) 20% (C) 25% (D) 30%

Q55. If $3x + 5 = 20$, then x equals

(A) 5 (B) 15 (C) 7 (D) 3

Q56. An item marked at ₹800 is sold at a discount of 15%. The selling price is

(A) ₹640 (B) ₹720 (C) ₹700 (D) ₹680

- Q57.** The compound interest on ₹10,000 at 10% per annum for 2 years, compounded annually, is
(A) ₹2,000 (B) ₹2,200 (C) ₹1,900 (D) ₹2,100
- Q58.** 12.5% expressed as a fraction is
(A) $\frac{1}{16}$ (B) $\frac{1}{5}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$
- Q59.** A man covers half of a journey at 30 km/h and the other half at 60 km/h. His average speed for the whole journey is
(A) 50 km/h (B) 42 km/h (C) 45 km/h (D) 40 km/h
- Q60.** A man travels 120 km at 40 km/h and returns along the same route at 60 km/h. His average speed for the whole journey is
(A) 50 km/h (B) 52 km/h (C) 45 km/h (D) 48 km/h
- Q61.** The perimeter of a rectangle of length 12 cm and breadth 8 cm is
(A) 96 cm (B) 40 cm (C) 20 cm (D) 48 cm
- Q62.** A train 200 m long crosses a platform 300 m long at 20 m/s. The time taken is
(A) 15 seconds (B) 10 seconds (C) 30 seconds (D) 25 seconds
- Q63.** A 20-litre solution contains 25% milk. The quantity of milk in it is
(A) 4 litres (B) 5 litres (C) 8 litres (D) 6 litres
- Q64.** The average marks of three students is 60. Two of them scored 55 and 65. The third scored
(A) 65 (B) 55 (C) 62 (D) 60
- Q65.** The simple interest on ₹5,000 at 8% per annum for 3 years is
(A) ₹1,400 (B) ₹1,000 (C) ₹1,200 (D) ₹1,600
- Q66.** Two fair coins are tossed together. The probability of getting at least one head is
(A) $\frac{1}{4}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{3}{4}$
- Q67.** An article costing ₹500 is sold at a profit of 20%. The selling price is
(A) ₹620 (B) ₹520 (C) ₹600 (D) ₹650
- Q68.** A can complete a piece of work in 12 days and B in 18 days. Working together they will finish it in
(A) 6 days (B) 7.5 days (C) 7.2 days (D) 9 days
- Q69.** A train 150 m long, running at 54 km/h, crosses a pole in
(A) 8 seconds (B) 10 seconds (C) 12 seconds (D) 15 seconds
- Q70.** The circumference of a circle is 44 cm. Its radius ($\pi = \frac{22}{7}$) is
(A) 14 cm (B) 7 cm (C) 3.5 cm (D) 21 cm
- Q71.** Two numbers are in the ratio 3 : 5 and their sum is 64. The smaller number is
(A) 40 (B) 27 (C) 21 (D) 24
- Q72.** The fraction $\frac{7}{8}$ expressed as a percentage is
(A) 78% (B) 85% (C) 87.5% (D) 88.5%
- Q73.** If 6 men complete a task in 10 days, 4 men will complete it in
(A) 16 days (B) 12 days (C) 20 days (D) 15 days
- Q74.** The average of 5 numbers is 20. If the number 30 is removed, the average of the remaining numbers is
(A) 17.5 (B) 18 (C) 16 (D) 19
- Q75.** In an examination 30% of the students failed and 350 students passed. The total number of students is
(A) 600 (B) 550 (C) 500 (D) 450

SECTION D — GENERAL ENGLISH

Directions (Q76-Q77): Read the following passage and answer the questions that follow.

A written examination can measure preparation but not persistence. Employers who rely on it alone accept a known limitation in exchange for fairness, since a test administered identically to everyone removes the influence of connection and privilege. The trade-off is deliberate rather than careless.

Q76. The chief advantage of a written examination, according to the passage, is
(A) its impartiality (B) its low cost (C) its brevity (D) that it measures persistence

Q77. The phrase “the trade-off is deliberate” means that the limitation is
(A) accidental (B) knowingly accepted (C) unavoidable by law (D) widely unknown

Directions (Q78-Q79): Read the following passage and answer the questions that follow.

Railway capacity, not mine capacity, has often been the binding constraint on coal despatch in India. A mine can raise production within a season, but a new siding or a doubled line takes years to commission. Stockpiles at the pithead are therefore less a sign of surplus than of a bottleneck further down the chain.

Q78. According to the passage, large pithead stockpiles indicate
(A) falling demand for coal (B) a transport bottleneck (C) poor coal quality (D) excess mining licences

Q79. In the passage, “binding constraint” refers to
(A) the factor that actually limits output (B) a legal contract (C) a temporary shortage of labour (D) a financial penalty

Directions (Q80-Q81): Read the following passage and answer the questions that follow.

Rehabilitation of land after opencast mining is frequently described as an afterthought, but the sequence of work decides the outcome. Topsoil that is stored separately at the start can be replaced later to support plantation; topsoil that is mixed with overburden is effectively lost. Restoration, in other words, begins on the first day of excavation.

- Q80.** The passage argues that successful land restoration depends chiefly on
(A) the size of the mine (B) planning from the beginning of excavation (C) the quantity of rainfall (D) the number of workers employed
- Q81.** The phrase “an afterthought” is used to indicate something
(A) carefully planned (B) technically impossible (C) legally required (D) considered too late
- Q82.** Fill in the blank: “The train ___ before we reached the station.”
(A) had left (B) was leaving (C) is leaving (D) has left
- Q83.** Choose the correct improvement: “Each of the students have submitted the form.”
(A) have been submitted (B) are submitting (C) has submitted (D) No improvement
- Q84.** Choose the word most opposite in meaning to TRANSPARENT.
(A) Clear (B) Opaque (C) Fragile (D) Visible
- Q85.** The idiom “to let the cat out of the bag” means
(A) to make a false promise (B) to escape danger (C) to reveal a secret (D) to waste an opportunity
- Q86.** One who is unable to read or write is called
(A) Illegal (B) Illegible (C) Ignorant (D) Illiterate
- Q87.** Fill in the blank: “He insisted ___ going there alone.”
(A) for (B) with (C) to (D) on
- Q88.** Choose the word most opposite in meaning to PERMANENT.
(A) Lasting (B) Temporary (C) Stable (D) Fixed
- Q89.** Choose the correctly spelt word.
(A) Privilage (B) Priviledge (C) Privelege (D) Privilege
- Q90.** Identify the part containing an error: “The teacher gave / the students / many advices. / No error”
(A) The teacher gave (B) many advices (C) the students (D) No error
- Q91.** Fill in the blank: “He has been living here ___ 2015.”
(A) by (B) from (C) since (D) for
- Q92.** Identify the part containing an error: “He don’t know / the answer / to this question. / No error”
(A) to this question (B) He don’t know (C) the answer (D) No error
- Q93.** Identify the part containing an error: “I have visited / Delhi / last year. / No error”
(A) I have visited (B) last year (C) Delhi (D) No error
- Q94.** Fill in the blank: “He is ___ honest man.”
(A) the (B) an (C) a (D) no article
- Q95.** The idiom “to hit the nail on the head” means
(A) to make a serious mistake (B) to work with tools (C) to act violently (D) to say exactly the right thing
- Q96.** Fill in the blank: “She is good ___ mathematics.”
(A) on (B) at (C) in (D) with
- Q97.** Identify the part containing an error: “He returned back / from Delhi / yesterday evening. / No error”
(A) He returned back (B) yesterday evening (C) from Delhi (D) No error
- Q98.** One who loves mankind and works for its welfare is called a
(A) philatelist (B) misanthrope (C) philosopher (D) philanthropist
- Q99.** Choose the word most similar in meaning to ABANDON.
(A) Repair (B) Forsake (C) Acquire (D) Protect
- Q100.** Choose the correct improvement: “He prefers coffee than tea.”
(A) coffee to tea (B) coffee over than tea (C) coffee from tea (D) No improvement
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SECTION 1 — ENGINEERING MATHEMATICS

- Q101.** The number of elements in the power set of a set with 5 elements is
(A) 64 (B) 10 (C) 32 (D) 25
- Q102.** A square matrix A is singular if
(A) $\det(A) \neq 0$ (B) A is orthogonal (C) A is symmetric (D) $\det(A) = 0$
- Q103.** The rank of the matrix $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is
(A) 0 (B) 2 (C) not defined (D) 1
- Q104.** A and B are independent events with $P(A) = 0.4$ and $P(B) = 0.5$. Then $P(A \cup B)$ is
(A) 0.90 (B) 0.70 (C) 0.60 (D) 0.20
- Q105.** The variance of a random variable that takes a single constant value with probability 1 is
(A) undefined (B) 1 (C) 0 (D) equal to the mean
- Q106.** If A is a 3×3 matrix with $\det(A) = 2$, then $\det(2A)$ is
(A) 16 (B) 32 (C) 8 (D) 4
- Q107.** The value of the definite integral $\int_0^1 x \, dx$ is
(A) 1/2 (B) 1/3 (C) 2 (D) 1
- Q108.** If $|A| = 3$ and $|B| = 4$, the number of functions from A to B is
(A) 24 (B) 12 (C) 81 (D) 64
- Q109.** For the recurrence $a_n = a_{n-1} + 2$ with $a_1 = 1$, the value of a_{10} is
(A) 21 (B) 17 (C) 20 (D) 19
- Q110.** A bag contains 3 red and 2 blue balls. Two balls are drawn one after another without replacement. The probability that both are red is
(A) 9/25 (B) 3/10 (C) 1/2 (D) 2/5
- Q111.** In a partially ordered set, the least upper bound of a pair of elements is called the
(A) infimum (B) complement (C) minimal element (D) supremum
- Q112.** For a Poisson distribution
(A) variance = mean² (B) mean < variance (C) mean = variance (D) mean > variance

SECTION 2 — DIGITAL LOGIC

- Q113.** The 1's complement of the binary number 0101 is
(A) 1010 (B) 1011 (C) 1101 (D) 0110
- Q114.** The number of select lines required for an 8-to-1 multiplexer is
(A) 2 (B) 4 (C) 8 (D) 3
- Q115.** The number of flip-flops required to build a 4-bit shift register is
(A) 16 (B) 2 (C) 4 (D) 8
- Q116.** The output of a 2-input XNOR gate is 1 when
(A) both inputs are 0 only (B) the inputs differ (C) both inputs are 1 only (D) the inputs are equal
- Q117.** The Gray code equivalent of the binary number 1010 is
(A) 1101 (B) 1010 (C) 0101 (D) 1111
- Q118.** The expression $A + A'B$ simplifies to
(A) $A + B$ (B) B (C) A (D) AB
- Q119.** A 2-input NOR gate with both inputs tied together behaves as
(A) an OR gate (B) an AND gate (C) an XOR gate (D) a NOT gate
- Q120.** The function $F(A, B) = A \oplus B$ expressed as a sum of minterms is
(A) $\sum m(2, 3)$ (B) $\sum m(1, 2)$ (C) $\sum m(0, 1)$ (D) $\sum m(0, 3)$

SECTION 3 — COMPUTER ORGANIZATION & ARCHITECTURE

- Q121.** A 2-way set-associative cache contains 128 blocks in total. The number of sets is
(A) 16 (B) 64 (C) 256 (D) 128
- Q122.** Which I/O technique requires the least CPU involvement per word transferred?
(A) Interrupt-driven I/O (B) Programmed I/O (C) Polling (D) DMA
- Q123.** Zero-address instructions are used in machines organized around
(A) memory-to-memory operations (B) an accumulator (C) general-purpose registers (D) a stack
- Q124.** In a system with a two-level page table and no TLB hit, the number of memory accesses needed to read one data word is
(A) 2 (B) 4 (C) 1 (D) 3
- Q125.** The program counter is normally incremented during the
(A) execute phase of an ALU operation (B) operand decode phase (C) fetch phase (D) interrupt phase only

- Q126.** In a two-address instruction format, the result of an operation is normally placed in
(A) one of the operand locations (B) a dedicated result register only (C) the program counter (D) main memory always
- Q127.** A distinguishing feature of a synchronous bus is
(A) a common clock shared by all devices (B) fully variable transfer timing (C) handshaking signals for every transfer (D) the absence of control lines
- Q128.** A pipeline stall caused by a conditional branch instruction is an example of
(A) control hazard (B) structural hazard (C) data hazard (D) no hazard
- Q129.** A cache has an access time of 10 ns and main memory 100 ns. The hit ratio is 0.95 and on a miss only main memory is accessed. The average memory access time is
(A) 55 ns (B) 14.5 ns (C) 19 ns (D) 10 ns
- Q130.** The cache mapping scheme in which a memory block may be placed in any cache line is
(A) fully associative (B) set associative (C) sector mapped (D) direct mapped
- Q131.** In a little-endian machine, the least significant byte of a multi-byte word is stored at
(A) the lowest address (B) a separate segment (C) the stack top (D) the highest address
- Q132.** A 5-stage pipeline has a stage delay of 2 ns and no hazards. The time to execute 100 instructions is
(A) 210 ns (B) 208 ns (C) 1000 ns (D) 200 ns

SECTION 4 — PROGRAMMING & DATA STRUCTURES

- Q133.** A queue implemented in a linear array with simple front and rear indices suffers from
(A) loss of FIFO order (B) false overflow even when slots are free (C) inability to store integers (D) $O(n)$ enqueue time
- Q134.** The most efficient data structure for implementing a priority queue supporting $O(\log n)$ insert and extract-min is
(A) a stack (B) a sorted array (C) a binary heap (D) an unsorted linked list
- Q135.** Which data structure is used for the evaluation of a postfix expression?
(A) Queue (B) Priority queue (C) Circular queue (D) Stack
- Q136.** In a threaded binary tree, NULL pointers are used to
(A) point to the in-order predecessor or successor (B) reduce the height of the tree (C) balance the tree (D) store additional data
- Q137.** Iterative reversal of a singly linked list requires a minimum of how many pointer variables?
(A) 1 (B) 4 (C) 2 (D) 3
- Q138.** Linear probing as a collision resolution technique suffers from
(A) chaining overhead (B) no clustering at all (C) secondary clustering (D) primary clustering
- Q139.** The minimum number of nodes in an AVL tree of height 3 (root at height 0) is
(A) 6 (B) 8 (C) 15 (D) 7
- Q140.** The time to search for a key in a height-balanced binary search tree with n nodes is
(A) $O(n)$ (B) $O(\log n)$ (C) $O(n \log n)$ (D) $O(1)$
- Q141.** The number of distinct binary search trees that can be formed with 3 distinct keys is
(A) 5 (B) 6 (C) 3 (D) 9
- Q142.** The postorder traversal of an expression tree produces the expression in
(A) prefix form (B) infix form (C) postfix form (D) level order
- Q143.** The height of a complete binary tree with n nodes (root at height 0) is
(A) $n/2$ (B) $\lceil n/2 \rceil$ (C) $\lceil \log_2 n \rceil$ (D) $\log_2(n + 1)$
- Q144.** The worst-case time to insert a node at the beginning of a singly linked list with n nodes is
(A) $O(n)$ (B) $O(n \log n)$ (C) $O(1)$ (D) $O(\log n)$

SECTION 5 — ALGORITHMS

- Q145.** Dijkstra's single-source shortest path algorithm may produce incorrect results when the graph has
(A) more than one source (B) negative edge weights (C) parallel edges (D) cycles
- Q146.** Which algorithm computes single-source shortest paths correctly in a graph having negative edge weights but no negative cycle?
(A) Bellman-Ford algorithm (B) Kruskal's algorithm (C) Dijkstra's algorithm (D) Prim's algorithm
- Q147.** The fractional knapsack problem is optimally solved by
(A) branch and bound (B) backtracking (C) the greedy method (D) dynamic programming
- Q148.** The time complexity of building a binary heap from an unsorted array of n elements using the bottom-up procedure is
(A) $O(n^2)$ (B) $O(n \log n)$ (C) $O(\log n)$ (D) $O(n)$
- Q149.** In a hash table with n keys, m slots and collision resolution by separate chaining, the average time for an unsuccessful search is
(A) $O(1 + n/m)$ (B) $O(1)$ (C) $O(n)$ (D) $O(\log n)$
- Q150.** The Floyd-Warshall all-pairs shortest path algorithm has time complexity
(A) $O(V^2)$ (B) $O(E \log V)$ (C) $O(VE)$ (D) $O(V^3)$
- Q151.** Binary search cannot be applied efficiently to a sorted singly linked list because
(A) comparisons are costlier (B) recursion is not allowed (C) accessing the middle element takes $O(n)$ time (D) the list is unsorted
- Q152.** Which of the following functions grows the fastest asymptotically?
(A) $n \log n$ (B) n^2 (C) 2^n (D) $n!$

Q153. Kruskal's algorithm for minimum spanning tree is based on which design technique?

(A) Backtracking (B) Dynamic programming (C) Greedy (D) Divide and conquer

Q154. The greedy activity-selection algorithm obtains an optimal solution by repeatedly choosing the activity with the

(A) highest value (B) shortest duration (C) earliest finishing time (D) earliest starting time

Q155. The worst-case time complexity of heapsort is

(A) $O(n \log n)$ (B) $O(n^2)$ (C) $O(\log n)$ (D) $O(n)$

Q156. Huffman coding is

(A) a greedy but sub-optimal algorithm (B) a greedy algorithm that produces an optimal prefix code (C) a dynamic programming algorithm producing fixed-length codes (D) a divide-and-conquer algorithm

SECTION 6 — THEORY OF COMPUTATION

Q157. Which of the following containments is correct?

(A) Regular $\subset$ CFL $\subset$ Recursive (B) Regular = CFL (C) Recursive $\subset$ CFL (D) CFL $\subset$ Regular

Q158. Regarding the classes of languages accepted by DFA and NFA

(A) the two classes are incomparable (B) both accept exactly the same class (C) DFAs accept strictly more languages (D) NFAs accept strictly more languages

Q159. The minimum number of states in a DFA over $\{0, 1\}$ accepting all strings ending with 01 is

(A) 4 (B) 2 (C) 3 (D) 5

Q160. The language of all palindromes over $\{a, b\}$ is

(A) not context-free (B) regular (C) context-free but not regular (D) finite

Q161. The tape of a standard Turing machine is

(A) read-only (B) unbounded in length (C) finite and fixed (D) write-only

Q162. The minimum number of states in a DFA accepting the language of the regular expression $(a + b)^*abb$ is

(A) 5 (B) 4 (C) 6 (D) 3

Q163. The language $L = \{w \in \{a, b\}^* \mid w \text{ has an equal number of } a\text{'s and } b\text{'s}\}$ is

(A) not context-free (B) finite (C) context-free but not regular (D) regular

Q164. The minimum number of states in a DFA over $\{a, b\}$ that accepts all strings in which the number of a 's is divisible by 3 is

(A) 4 (B) 2 (C) 6 (D) 3

SECTION 7 — COMPILER DESIGN

Q165. FIRST and FOLLOW sets are computed for the purpose of

(A) register allocation (B) constructing an LL(1) parsing table (C) peephole optimization (D) lexical analysis

Q166. Register allocation and assignment is performed during

(A) code generation (B) lexical analysis (C) semantic analysis (D) parsing

Q167. Moving a computation that does not change across iterations out of a loop is called

(A) constant folding (B) loop-invariant code motion (C) strength reduction (D) dead code elimination

Q168. The output of the lexical analysis phase is

(A) a stream of tokens (B) intermediate code (C) a parse tree (D) a symbol table only

Q169. A basic block in intermediate code is a sequence of instructions with

(A) a single entry point and a single exit point (B) at least one loop (C) multiple entry points (D) exactly one instruction

Q170. Which of the following is a machine-independent optimization?

(A) Instruction selection (B) Peephole optimization (C) Common sub-expression elimination (D) Register allocation

Q171. Handle pruning is a technique associated with

(A) top-down parsing (B) semantic analysis (C) bottom-up parsing (D) lexical analysis

Q172. The minimum number of temporary variables needed in the three-address code for $a = b \setminus * c + d \setminus * e$ is

(A) 1 (B) 4 (C) 3 (D) 2

SECTION 8 — OPERATING SYSTEM

Q173. With a page size of 1 KB and a 16-bit logical address, the number of pages in the logical address space is

(A) 16 (B) 1024 (C) 64 (D) 32

Q174. Which of the following is NOT one of the four necessary conditions for deadlock?

(A) Mutual exclusion (B) Circular wait (C) Hold and wait (D) Priority inversion

Q175. A process sitting in the ready queue is waiting for

(A) the CPU (B) a semaphore signal (C) a page to be loaded (D) an I/O device

Q176. External fragmentation is a problem primarily in

(A) variable-partition contiguous allocation (B) demand paging (C) TLB management (D) paging

Q177. The Translation Lookaside Buffer (TLB) is

(A) part of secondary storage (B) a page replacement policy (C) a cache of recently used page table entries (D) a file allocation index

- Q178.** Spinlocks are the preferred synchronization mechanism when
(A) critical sections are very long (B) critical sections are very short and multiple processors are available (C) there is a single CPU and long critical sections (D) processes never contend for the lock
- Q179.** The convoy effect is associated with which scheduling algorithm?
(A) FCFS (B) SJF (C) Multilevel feedback queue (D) Round Robin
- Q180.** Which of the following is NOT a requirement of a correct critical-section solution?
(A) Bounded waiting (B) Mutual exclusion (C) Progress (D) Assumptions about relative process speeds
- Q181.** In UNIX/Linux, the system call used to create a new process is
(A) exec() (B) wait() (C) fork() (D) pipe()
- Q182.** Which CPU scheduling algorithm gives the minimum average waiting time for a given set of processes arriving together?
(A) Priority (B) Shortest Job First (C) Round Robin (D) FCFS
- Q183.** For the reference string 1, 2, 3, 4, 1, 2, 5 with 3 frames and FIFO replacement, the number of page faults is
(A) 6 (B) 7 (C) 5 (D) 4
- Q184.** The optimal (OPT/MIN) page replacement algorithm
(A) gives the lowest possible fault rate but is not implementable in practice (B) suffers from Belady's anomaly (C) behaves identically to FIFO (D) is easily implementable
-

SECTION 9 — DATABASES

- Q185.** Two-phase locking (2PL) protocol guarantees
(A) deadlock freedom (B) recoverability only (C) no starvation (D) conflict serializability
- Q186.** In a B+ tree index, pointers to actual data records are stored in
(A) leaf nodes only (B) the root only (C) internal nodes only (D) both internal and leaf nodes
- Q187.** In SQL, COUNT(column_name)
(A) is always equal to COUNT(*) (B) counts NULL values as well (C) is invalid syntax (D) ignores NULL values
- Q188.** A schedule is conflict serializable if and only if its precedence (serializability) graph
(A) is acyclic (B) is bipartite (C) is a complete graph (D) contains a cycle
- Q189.** If relation R has 5 tuples and relation S has 4 tuples, the number of tuples in $R \times S$ is
(A) 4 (B) 9 (C) 5 (D) 20
- Q190.** The projection operation in relational algebra
(A) retains duplicate tuples (B) selects rows only (C) always returns one tuple (D) eliminates duplicate tuples
- Q191.** In SQL, the clause used to filter groups produced by GROUP BY is
(A) ORDER BY (B) HAVING (C) WHERE (D) DISTINCT
- Q192.** Which SQL set operator eliminates duplicate rows by default?
(A) CROSS JOIN (B) UNION ALL (C) JOIN (D) UNION
-

SECTION 10 — COMPUTER NETWORKS

- Q193.** Which DNS resource record maps a domain name to an IPv4 address?
(A) PTR (B) CNAME (C) A (D) MX
- Q194.** The number of /27 subnets that can be carved out of a single /24 network is
(A) 4 (B) 32 (C) 8 (D) 16
- Q195.** The number of usable host addresses in a /30 subnet is
(A) 2 (B) 4 (C) 1 (D) 6
- Q196.** The layer responsible for reliable end-to-end (process-to-process) delivery is the
(A) network layer (B) transport layer (C) data link layer (D) session layer
- Q197.** The correct sequence of segments in the TCP connection establishment handshake is
(A) ACK, SYN, SYN+ACK (B) SYN, SYN+ACK, ACK (C) SYN, FIN, ACK (D) SYN, ACK, SYN
- Q198.** A LAN switch operates primarily at which OSI layer?
(A) Network (B) Data link (C) Physical (D) Transport
- Q199.** A sliding window protocol with a sender window size of 1 reduces to
(A) Stop-and-Wait (B) Go-Back-N (C) Selective Repeat (D) piggybacked sliding window
- Q200.** The default subnet mask of a Class B IPv4 network is
(A) 255.0.0.0 (B) 255.255.0.0 (C) 255.255.255.0 (D) 255.255.255.255
-

ANSWER KEY — MOCK CBT 2

Paper-I (Q1 - Q100)

1-B 2-B 3-D 4-C 5-D 6-D 7-A 8-A 9-C 10-C
11-D 12-A 13-D 14-A 15-C 16-B 17-B 18-B 19-D 20-A
21-C 22-C 23-B 24-D 25-A 26-A 27-A 28-A 29-C 30-B
31-A 32-B 33-B 34-D 35-C 36-C 37-A 38-C 39-D 40-D
41-D 42-A 43-A 44-C 45-A 46-D 47-A 48-A 49-A 50-D
51-D 52-B 53-C 54-C 55-A 56-D 57-D 58-D 59-D 60-D
61-B 62-D 63-B 64-D 65-C 66-D 67-C 68-C 69-B 70-B
71-D 72-C 73-D 74-A 75-C 76-A 77-B 78-B 79-A 80-B
81-D 82-A 83-C 84-B 85-C 86-D 87-D 88-B 89-D 90-B
91-C 92-B 93-A 94-B 95-D 96-B 97-A 98-D 99-B 100-A

Paper-II (Q101 - Q200)

101-C 102-D 103-D 104-B 105-C 106-A 107-A 108-D 109-D 110-B
111-D 112-C 113-A 114-D 115-C 116-D 117-D 118-A 119-D 120-B
121-B 122-D 123-D 124-D 125-C 126-A 127-A 128-A 129-B 130-A
131-A 132-B 133-B 134-C 135-D 136-A 137-D 138-D 139-D 140-B
141-A 142-C 143-C 144-C 145-B 146-A 147-C 148-D 149-A 150-D
151-C 152-D 153-C 154-C 155-A 156-B 157-A 158-B 159-C 160-C
161-B 162-B 163-C 164-D 165-B 166-A 167-B 168-A 169-A 170-C
171-C 172-C 173-C 174-D 175-A 176-A 177-C 178-B 179-A 180-D
181-C 182-B 183-B 184-A 185-D 186-A 187-D 188-A 189-D 190-D
191-B 192-D 193-C 194-C 195-A 196-B 197-B 198-B 199-A 200-B

Score sheet

Paper-I: ____ / 100 (qualifying: 40 UR/EWS, 35 OBC-NCL, 30 SC/ST/PwD)

Paper-II: ____ / 100 (same qualifying marks)

Total: ____ / 200

Section-wise wrong answers — GA ____ Reasoning ____ Numerical ____ English ____ | Engg Maths ____ Digital Logic ____ COA ____ Prog & DS ____
Algorithms ____ TOC ____ Compilers ____ OS ____ DBMS ____ CN ____